Informatica

Informatica® Cloud Data Integration
Spring 2018

Hadoop Connector

Informatica Cloud Data Integration Hadoop Connector
Spring 2018
July 2018

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Preface

Use Hadoop Connector to learn how to read from or write to the Hadoop clusters by using Cloud Data Integration. Learn to create a connection, and develop and run synchronization tasks in Cloud Data Integration.

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CHAPTER 1

Introduction to Hadoop Connector

This chapter includes the following topics:

- Hadoop Connector Overview, 6
- Supported Objects and Task Operations, 7
- Hadoop Description, 7

Hadoop Connector Overview

The Informatica Hadoop connector allows you to perform the Read and Insert operations on Hadoop. You can use the Hadoop connector on a kerberos cluster and a non-kerberos cluster.

Hadoop Connector on a Kerberos Cluster

On a kerberos cluster, you can install the Secure Agent outside the cluster and perform read and write operations successfully. The Hadoop connector exhibits the following features on a kerberos cluster:

- Connects to Hive to perform relevant data operations.
- Supports all operators supported in HiveQL.
- Supports the AND conjunction between filters. It supports both AND and OR conjunctions in advanced filters.
- Supports filtering on all filterable columns in Hive tables.

The Hadoop connector supports the following distributions on a kerberos cluster:

- Cloudera 5.8 - 5.13
- Hortonworks 2.5

Hadoop Connector on a non-Kerberos Cluster

On a non-kerberos cluster, you can perform the read operation if the Secure Agent is installed outside the cluster. However, you must install the Secure Agent on one of the nodes of the Hadoop cluster to perform the write operation successfully. The Hadoop connector exhibits the following features on a non-kerberos cluster:

- Connects to Hive and Cloudera Impala to perform relevant data operations.
- Supports all operators supported in HiveQL.
- Supports the AND conjunction between filters. It supports both AND and OR conjunctions in advanced filters.

- Supports filtering on all filterable columns in Hive and Impala tables.

The Hadoop connector supports the following distributions on a non-kerberos cluster:

- Amazon EMR
- Cloudera 5.0
- Hortonworks 2.1
- MapR 3.1
- Pivotal 2.0

Supported Objects and Task Operations

The following table displays the list of objects and task operations supported by Hadoop Connector. You can read data from or write data to partitioned table in Hadoop.

Objects	Read	Insert	Data Preview
Tables in Hive (kerberos and non-kerberos clusters)	Yes	Yes	Yes
Tables in Impala (non-kerberos only)	Yes	No	Yes

Hadoop Description

The Apache Hadoop develops open source software for reliable, scalable, and distributed computing.

The Apache Hadoop software library is a framework that allows for the distributed processing of large data sets across clusters of computers using simple programming models. It is designed to scale up from single servers to thousands of machines, each offering local computation and storage. Rather than rely on hardware to deliver high-availability, the library itself is designed to detect and handle failures at the application layer, so delivering a highly-available service on top of a cluster of computers, each of which may be prone to failures.

The project includes these modules:

- Hadoop Common: The common utilities that support the other Hadoop modules.
- Hadoop Distributed File System (HDFS): A distributed file system that provides high-throughput access to application data.
- Hadoop YARN: A framework for job scheduling and cluster resource management.
- Hadoop MapReduce: A YARN-based system for parallel processing of large data sets.

Other Hadoop-related projects at Apache include:

- Ambari™: A web-based tool that provisions, manages, and monitors Apache Hadoop clusters which include support for Hadoop HDFS, Hadoop MapReduce, Hive, HCatalog, HBase, ZooKeeper, Oozie, Pig and Sqoop. Ambari also provides a dashboard for viewing cluster health such as heatmaps and ability to view MapReduce, Pig and Hive applications visually along with features to diagnose their performance characteristics in a user-friendly manner.

- Avro™: A data serialization system.
- Cassandra™: A scalable multi-master database with no single points of failure.
- Chukwa™: A data collection system for managing large distributed systems.
- HBase™: A scalable, distributed database that supports structured data storage for large tables.
- Hive™: A data warehouse infrastructure that provides data summarization and ad hoc querying.
- Mahout™: A Scalable machine learning and data mining library.
- Pig™: A high-level data-flow language and execution framework for parallel computation.
- ZooKeeper™: A high-performance coordination service for distributed applications.

Hadoop cluster can use Kerberos authentication to verify user accounts. You can use Kerberos authentication with Data Integration, with the Hadoop cluster, or with both. Kerberos is a network authentication protocol which uses tickets to authenticate access to services and nodes in a network. Kerberos uses a Key Distribution Center to validate the identities of users and services and to grant tickets to authenticated user and service accounts. Users and services are known as principals. The Key Distribution Center has a database of principals and their associated secret keys that are used as proof of identity.

CHAPTER 2

Hadoop Connections

This chapter includes the following topics:

- Enabling Hadoop Connector, 9
- Hadoop connection properties, 10

Enabling Hadoop Connector

To enable Hadoop Connector, contact Informatica Global Customer Support. It usually takes 15 minutes for the connector to download to secure agent, after it is enabled.

The user must have appropriate read and write permissions on the temporary directory. For Linux, /tmp is the default directory. Make sure that the same temporary directory structure is available on the Secure Agent system and the HDFS cluster.

Within an organization, you cannot enable licenses for more than one Hadoop distribution. If you want to use multiple Hadoop distributions, you must create separate sub-organizations for each Hadoop distribution that you want to use and assign license for the Hadoop distribution. You can also switch from one Hadoop distribution license to another within a sub-organization. To switch from one Hadoop distribution license to another, you must perform the following tasks:

1. In Administrator, select **Organization**.
2. Click **Sub-Organizations**.
3. Select the sub-organization for which you want to switch the Hadoop distribution.
4. Click **Licenses**.
5. On the Custom Licenses section, unselect the license for the currently active Hadoop distribution.
6. Delete the existing Hadoop distribution package folder and the hadoop distribution `.checksum` file from the following location:
   `<Secure Agent Installation Directory>/downloads`
7. On the **Licenses** page, select the license for the Hadoop distribution package that you want to use for the sub-organization.
8. Restart the Secure Agent.

For a Kerberos Cluster

Perform the following tasks to use the connector on a Hadoop cluster:

- You must have the license for the `Hortonworks_2_5` or `Cloudera_5_8` packages for your organization. Contact Informatica Global Customer Support for more information about licensing.

- If the Secure Agent is installed outside the Hadoop cluster, complete the following steps:
  1. Make an entry of the Key Distribution Center (KDC) in the `/etc/hosts` file.
  2. Copy the `/etc/krb5.conf` file from the cluster node where the Hive service is running to the `<Secure Agent installation directory>/jre/lib/security` directory of the Secure Agent.
  3. Restart the Secure Agent.

Hadoop connection properties

To use Hadoop Connector in a synchronization task, you must configure the connection properties.

Important: Hadoop Connector is deprecated and has been moved to maintenance mode. Informatica intends to drop support in a future release. Informatica recommends that you use Hive Connector to access the Hadoop clusters.

The following table describes the Hadoop connection properties:

Connection property	Description
Username	The username of schema of Hadoop component.
Password	The password of schema of Hadoop component.
JDBC Connection URL	The JDBC URL to connect to the Hadoop Component. Refer "JDBC URL" on page 11
Driver	The JDBC driver class to connect to the Hadoop Component. For more information, see the Setting Hadoop Classpath for various Hadoop Distributions topic.
Commit Interval	The Batch size, in rows, to load data to hive.
Hadoop Installation Path	The Installation path of the Hadoop component. Not applicable to a kerberos cluster.
Hive Installation Path	Hive Installation Path Not applicable to a kerberos cluster.
HDFS Installation Path	The HDFS Installation Path. Not applicable to a kerberos cluster.
HBase Installation Path	The HBase Installation Path. Not applicable to a kerberos cluster.
Impala Installation Path	The Impala Installation Path. Not applicable to a kerberos cluster.
Miscellaneous Library Path	The library that communicates with Hadoop. Not applicable to a kerberos cluster.

Connection property	Description
Enable Logging	Enable logging enables the log messages. **Note:** The Enable Logging connection parameter is place-holder for a future release, and its state has no impact on connector functionality.
Hadoop Distribution	The Hadoop distributions for which you can use Kerberos Authentication. You can use Kerberos authentication for the Cloudera and HDP Hadoop distributions.
Authentication Type	You can select native or Kerberos authentication.
Key Tab File	The file that contains encrypted keys and Kerberos principals to authenticate the machine.
Hive Site XML	The directory where the core-site.xml, hive-site.xml and hdfs-site.xml are located. The three XML files must locate in the same location.
Superuser Principle Name	Users assigned to the superuser privilege can perform all the tasks that a user with the administrator privilege can perform.
Impersonation Username	You can enable different users to run mappings in a Hadoop cluster that uses Kerberos authentication or connect to sources and targets that use Kerberos authentication. To enable different users to run mappings or connect to big data sources and targets, you must configure user impersonation.

Note: Installation paths are the paths where you place the Hadoop jar. Hadoop Connector loads the libraries from installation paths before it sends instructions to Hadoop. When you use Kerberos Authentication type, you need not specify the Hadoop installation path, Hive installation path, HDFS installation path, HBase Installation path, Impala installation path, and Miscellaneous Library path.

If you do not use Kerberos Authentication and do not mention the installation path, you can set the Hadoop classpath for Amazon EMR, HortonWorks, MapR and Cloudera.

When you perform an insert operation on non-Kerberos clusters, the Secure Agent uses the *hadoop fs -put <FS> <HDFS>* command to upload the file to the HDFS and uses the *hadoop fs -rm -r <HDFS>* command to delete the file from the HDFS. When you enable Kerberos authentication, the Secure Agent does not use the Hadoop commands to write data to or delete data from the HDFS.

JDBC URL

The connector connects to different components of Hadoop with JDBC. The URL format and parameters differ from components to components.

The Hive uses following JDBC URL format:

```
jdbc:<hive/hive2>://<server>:<port>/<schema>
```

The significance of URL parameters is discussed below:

- hive/hive2 : Contains the protocol information. The version of the Thrift Server, that is, hive for HiveServer and hive2 for HiveServer2.
- Server, port – server and port information of the Thrift Server.
- Schema – The hive schema which the connector needs to access.

For example, `jdbc:hive2://invrlx63iso7:10000/default` connects the default schema of Hive, uses a Hive Thrift server HiveServer2 that starts on the server invrlx63iso7 on port 10000.

Hadoop Connecter uses the Hive thrift server to communicate with Hive.

The command to start the Thrift server is –hive –service hiveserver2.

Cloudera Impala uses the JDBC URL in the following format:

`jdbc:hive2://<server>:<port>/;auth=<auth mechanism>`

JDBC Driver Class

The JDBC Driver class varies among Hadoop components. For example, `org.apache.hive.jdbc.HiveDriver` for Hive and Impala.

Setting Hadoop classpath for non-Kerberos clusters

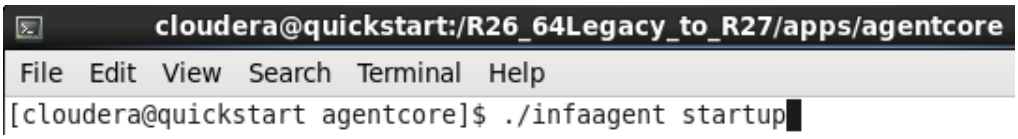
For non-Kerberos Hadoop clusters, if you do not mention the installation paths in connection properties, you can still perform the connection operations by setting the class path. You must simply set the classpath for the respective distributions.

This section helps you to set the classpath for Amazon EMR, HortonWorks, Pivotal, and MapR.

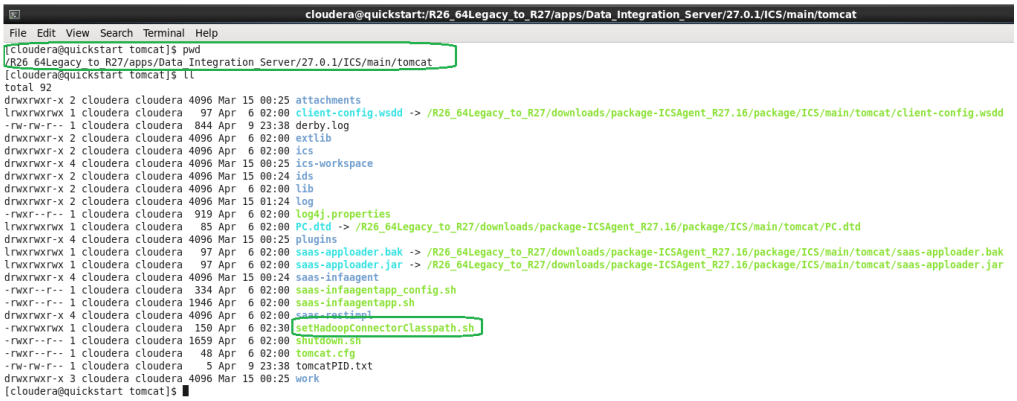
Setting Hadoop classpath for Amazon EMR, HortonWorks, Pivotal, and MapR

Perform the following steps to generate the `setHadoopConnectorClasspath.sh` for Amazon, Horton works, Pivotal, and MapR.

1. Use the command displayed in the following image to start the Secure Agent:

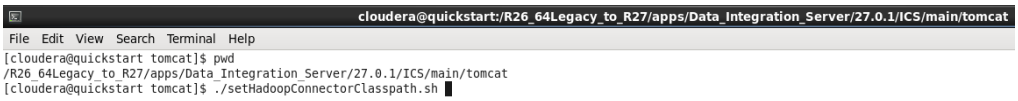


2. Create the Hadoop connection using the connector.
3. Test the connection. This generates the `setHadoopConnectorClasspath.sh` file in `<Secure Agent installation directory>/apps/Data_Integration_Server/<latest DIS version>/ICS/main/tomcat` path.

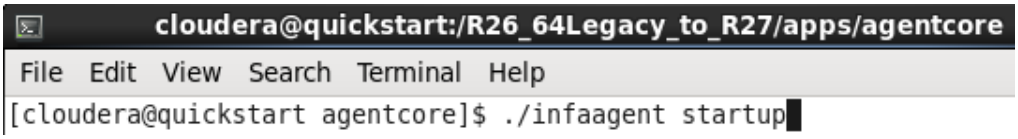


4. Stop the Secure Agent.

5. From `<Secure Agent installation directory>`, execute the `. ./main/tomcat/setHadoopConnectorClasspath.sh` command.



6. Start the Secure Agent and execute the synchronization tasks:



Note: If you want to generate the `setHadoopConnectorClasspath.sh` file again, then delete the existing classpath and regenerate.

In certain cases the Hadoop classpath may point to the incorrect classpath. In this case, you must direct the Hadoop classpath to the correct classpath.

a. Enter the command `hadoop classpath` from the terminal to display the stream of jars.

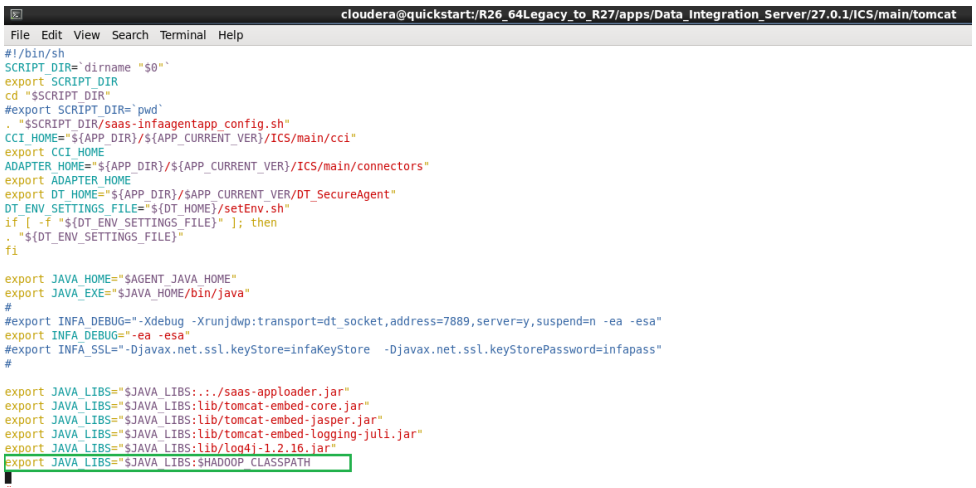
b. Copy the stream of jars in a notepad.

c. If the following entries are present in the notepad file, delete them:

- `:/opt/mapr/hadoop/hadoop-0.20.2/bin/../hadoop*core*.jar`
- `:/opt/mapr/hadoop/hadoop-0.20.2/bin/../lib/commons-logging-api-1.0.4.jar` (retain the latest version and delete the previous)

d. Copy the remaining content and export it to a variable called `HADOOP_CLASSPATH`.

e. In **saas-infaagentapp.sh** file make the following entry:



f. Restart the Secure Agent.

CHAPTER 3

Synchronization Tasks with Hadoop Connector

This chapter includes the following topics:

- Creating Hadoop Synchronization Task, 14
- Data Filters, 16
- Rules and Guidelines for Synchronization Tasks, 17

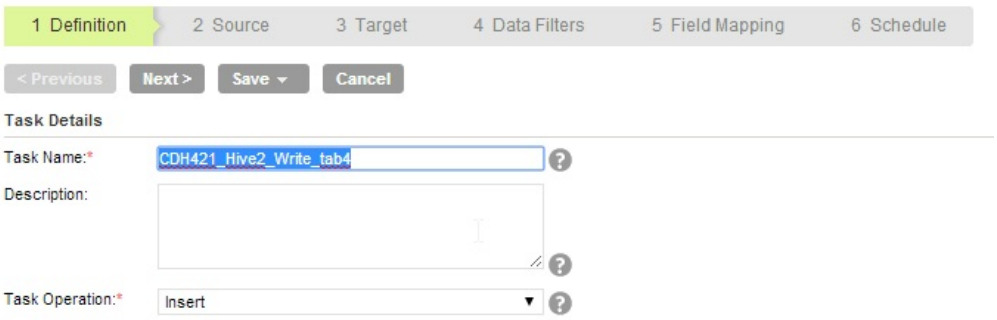
Creating Hadoop Synchronization Task

Note: You need to create a connection before getting started with a synchronization task.

The following steps help you to setup a synchronization task in Data Integration.

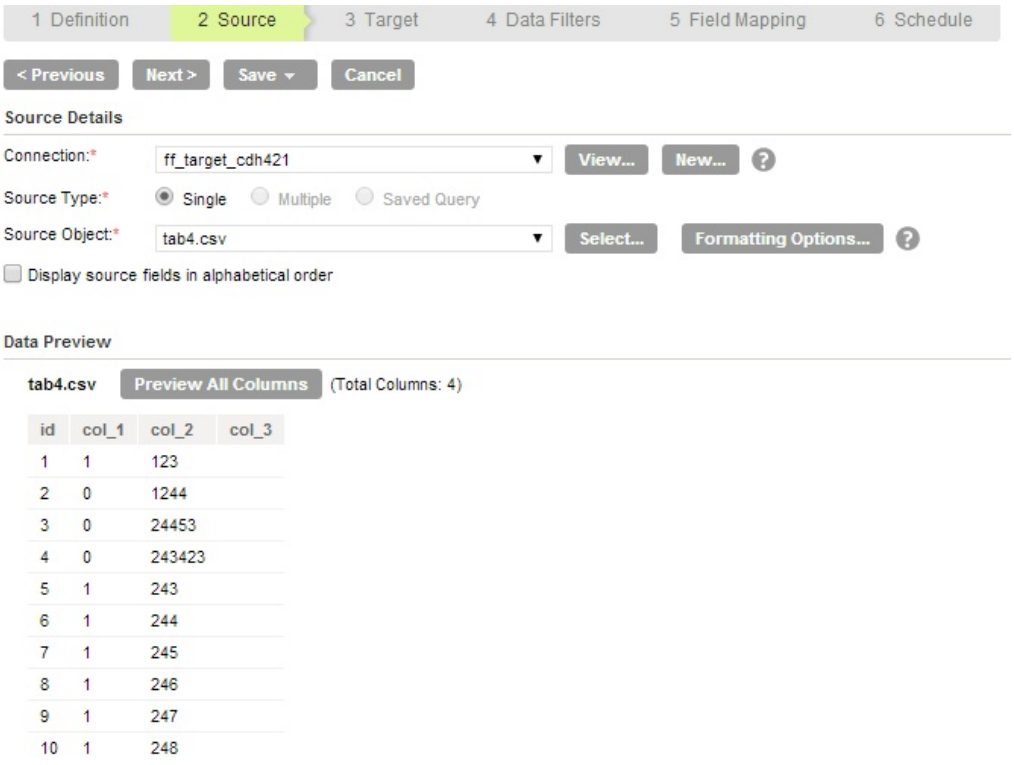
Let us consider the task operation Insert to perform the synchronization task.

1. Click **New** > **Tasks**.
2. Select **Synchronization Task** and click **Create**.
3. In the **New Synchronization Task** page, configure the task details in the **Definition** tab.



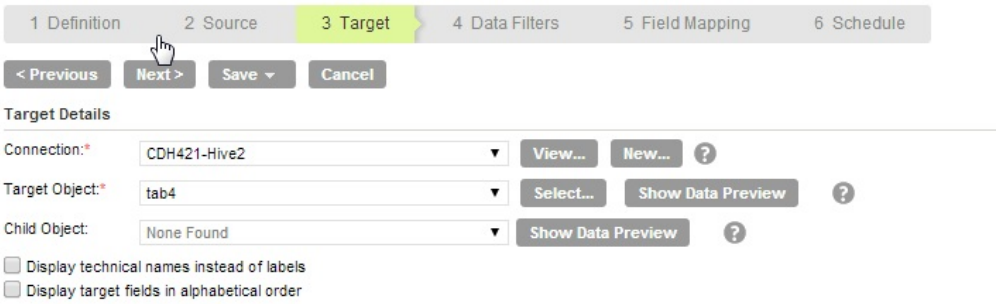
4. Specify the **Task Name**, provide a **Description**, and select Insert as the **Task Operation**.
5. Click **Next**.

The **Source** tab appears.



6. Select the source **Connection**, **Source Type**, and **Source Object** to be used for the task.
7. Click **Next**.

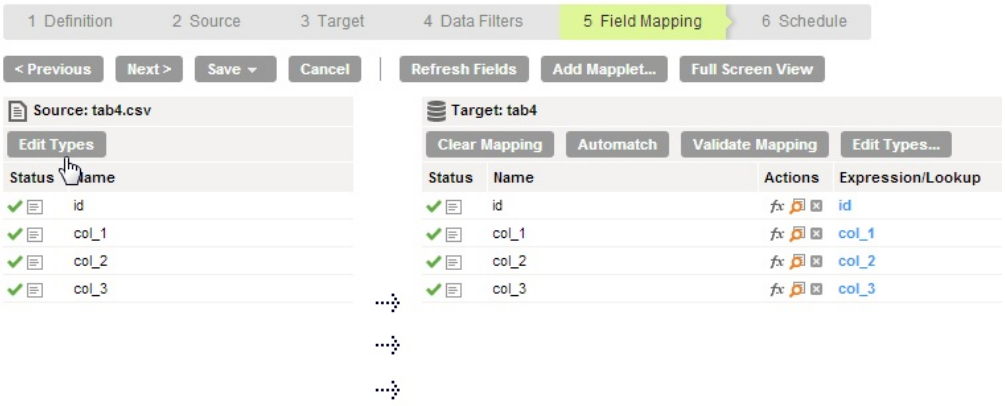
   The **Target** tab appears.
8. Select the target **Connection** and **Target Object** required for the task.



9. Click **Next**.
10. In the **Data Filters** tab, by default, **Process all rows** is chosen. It is mandatory to assign the **_FLT_URL_Input_Parameters_Config_File_Path** data filter in the synchronization task.

    For more information, see "Data Filters" on page 16.
11. Click **Next**.

12. In **Field Mapping** tab, map source fields to target fields accordingly.



13. Click **Next**.

    The **Schedule** tab appears.

14. In **Schedule** tab, you can schedule the task as per the requirement.
15. Click **Finish** to complete the task.

Data Filters

Data filters help you to fetch specific data based on the APIs configured in `Config.csv` file.

The synchronization task processes the data based on the filter field assigned.

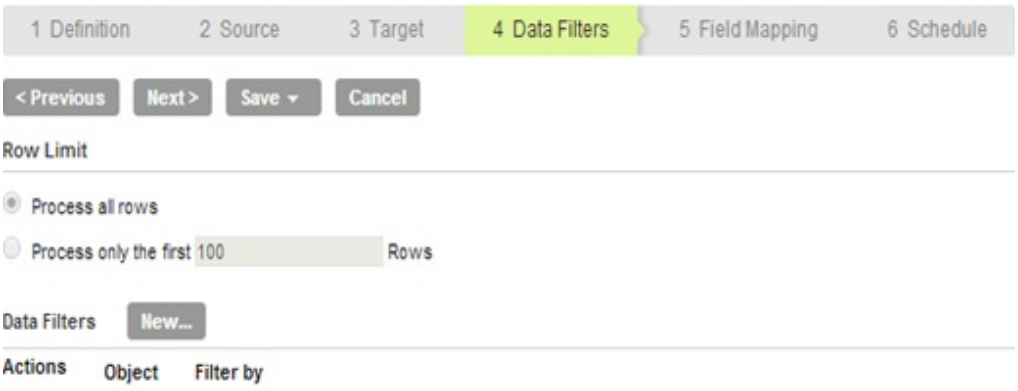
Note: Advanced data filters are not supported by Hadoop Connector.

Perform the following steps to configure data filters:

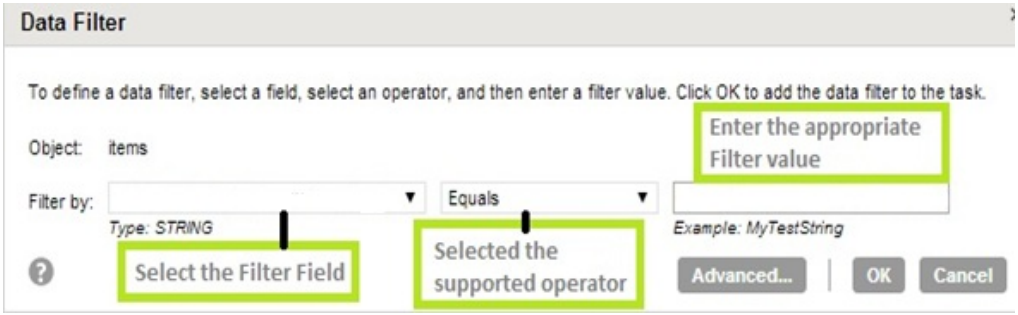
1. In the synchronization task, select **Data Filters** tab.

   The Data Filters tab appears.

2. Click **New** as shown in the following figure:



The Data Filter dialog box appears.



3. Specify the following details.

Field Type	Description
Object	Select the object for which you want to assign filter fields.
Filter By	Select the Filter Field.
Operator	Select the Equals operator. Only the Equals operator is supported with this release.
Filter Value	Enter the Filter value.

4. Click **OK**.

Rules and Guidelines for Synchronization Tasks

Consider the following rules and guidelines for synchronization tasks:

- When you use a Hadoop connection in a synchronization task to write data to a Hive table and the source table contains data of complex data type, the synchronization task fails.
- Ignore log4j initialization warnings in the session logs.

CHAPTER 4

Data Type Reference

This chapter includes the following topics:

- Data Type Reference Overview, 18
- Hadoop and Transformation Data Types, 18

Data Type Reference Overview

Data Integration uses the following data types in synchronization tasks, mapping tasks, and mappings with Hadoop:

Hadoop native data types

Hadoop data types appear in the Source transformations when you choose to edit metadata for the fields.

Transformation data types

Set of data types that appear in the transformations. They are internal data types based on ANSI SQL-92 generic data types, which the Secure Agent uses to move data across platforms. Transformation data types appear in all transformations in a mapping.

When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

Hadoop and Transformation Data Types

The following table lists the Hadoop data types that Data Integration supports and the corresponding transformation data types:

Hadoop Data Type	Transformation Data Type
String (255 length)	string
Integer	int, tinyint, smallint
Big Integer	bigint

Hadoop Data Type	Transformation Data Type
Float	float
Double	double
Boolean	boolean
Timestamp	timestamp

CHAPTER 5

Troubleshooting

This chapter includes the following topics:

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Increasing the Secure Agent memory

To increase performance and avoid runtime environment memory issues, perform the following steps:

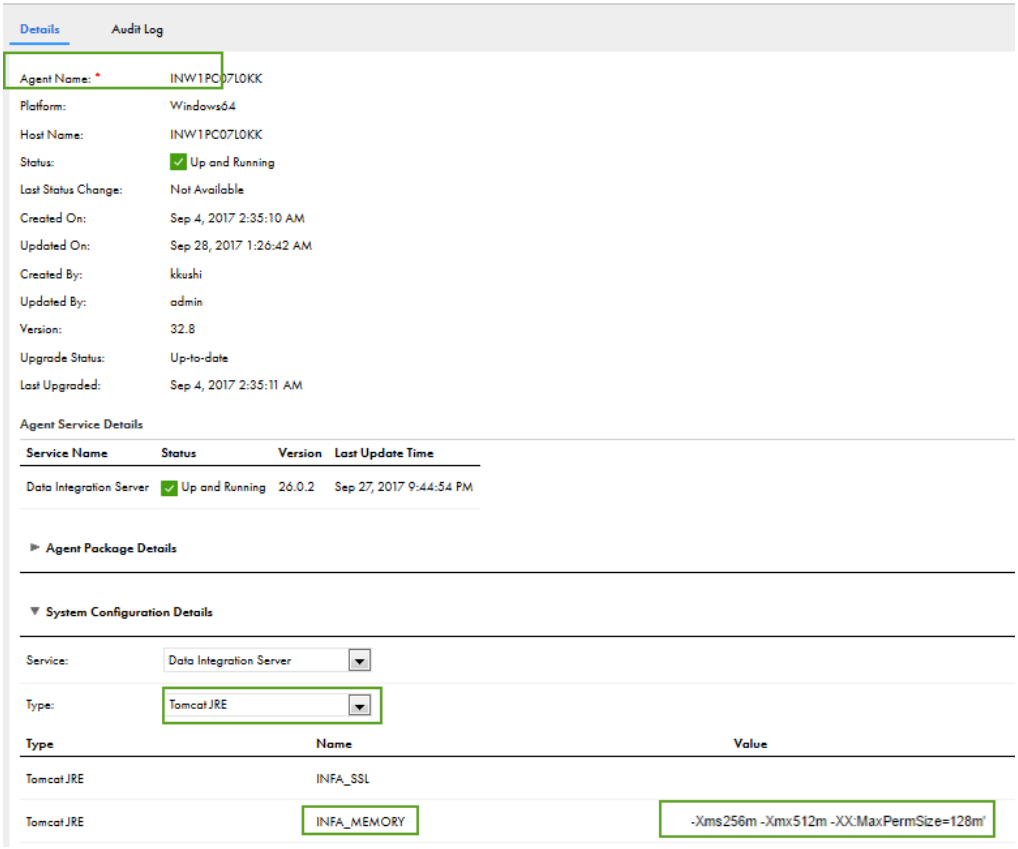
1. In Administrator, select **Runtime Environments**.
2. Select the Secure Agent for which you want to increase memory from the list of available Secure Agents.
3. In the upper-right corner, click **Edit**.
4. In the **System Configuration Details** section, select the **Type** as **DTM** for the Data Integration Service.
5. Edit **JVMOption1** as **-Xms4056m** and **JVMOption2** as **-Xmx4056m**.

   The following image shows the **Details** page:



6. In the **System Configuration Details** section, select the **Type** as **TomCatJRE**
7. Edit **INFA_memory** as **-Xms256m -Xmx512m**.

The following image shows the **Agent Details** page:



Note: The minimum and maximum values for the Java heap size are given as an example. Specify the size according to your requirements.

8. Restart the Secure Agent.

Additional Troubleshooting Tips

- When the connection is used as a target, the last batch of the insert load is not reflected in the record count. Refer the session logs for the record count of the last batch inserted.
- For example, if the commit interval is set to 1 million and the actual rows inserted are 1.1 million, the record count in the UI shows 1 million and the session logs reveal the row count of the reminder 100k records.
- Set the commit interval to the highest value possible before java.lang.OutofMemoryError is encountered.
- When the connection is used as a target to load data into Hadoop, ensure that all the fields are mapped.
- After a data load in Hive, Impala needs to be refreshed manually for the latest changes to the table to be reflected in Impala. In the current version, the connector does not automatically refresh Impala upon a Hive dataset insert.

- When you enable Apache Sentry authorization on a Cloudera cluster or Apache Ranger authorization on a Hortonworks cluster, you must enable the **UseExternalTableToWrite** property in the **Custom Configuration Details** section of the Secure Agent and restart the Secure Agent. Otherwise, the mapping tasks fails with the following error:
  `WRITER_1_*_1> JAVA PLUGIN_1762 [2018-03-13 11:15:17.884][ERROR]java.sql.SQLException: Error while processing statement: FAILED: Execution Error, return code 1 from` **`org.apache.hadoop.hive.ql.exec.MoveTask`**

Enabling the UseExternalTableToWrite Property

To enable the **UseExternalTableToWrite** property for the Secure Agent, perform the following steps:

1. In Administrator, select **Runtime Environments**.
2. Select the Secure Agent for which you want to set the custom configuration flag.
3. On the upper-right corner of the page, click **Edit**.
4. In the **Custom Configuration Details** section, select the **Service** as **Data Integration Server**.
5. Select the **Type** as **DTM**.
6. Add **UseExternalTableToWrite** in the **Name** field.
7. Set the **Value** as **True**.
8. Click the **Add** icon.
9. Click **Save**.
10. Restart the Secure Agent.

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